

Theo Hatzis, MSc

Physical Systems Expert specialising in Electronics (Analogue, Power, CPU), Battery Systems, and Semiconductor Devices. Extensive experience in Mixed-Signal/Precision AFE, BMS, Power Distribution Networks, and MedTech/Analytical Instrumentation (NMR, XRF, FIA, Sensors). Bridging innovation across Chemistry & Spectroscopy, Materials Science, and advanced Photonics (RF → Visible → X-Ray → Gamma). Driving cross-disciplinary solutions at the intersection of electronics, physical systems, RF, and advanced instrumental methods • Expertise & Consultancy

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I am an expert in Physical Systems and Electronics Hardware, with deep experience spanning electronics-adjacent batteries, MedTech, Scitech, and RF ware systems. I lead fast-paced technical projects in power electronics, battery management systems (BMS), and power delivery networks (PDN). I design mixed-signal and analogue front-end (AFE) circuits, and handle semiconductor validation, verification, and characterisation (VVC), including process, voltage, and temperature (PVT) assessments. My expertise extends to RF-aware systems and electronics-adjacent battery technologies, as well as MedTech and Scitech solutions. I apply this knowledge to tackle challenges across NMR, XRF, FIA, and various sensor platforms. My background covers Chemistry, Materials Science, Electrochemistry, Spectroscopy, and Real-Time Systems. I also bring skills in Control Theory, Robotics Control, Algorithms, and Neural Networks. This interdisciplinary approach enables me to deliver "Seniority-as-a-Service" for mission-critical hardware, challenges and bursts.

A veteran of the Bavarian/Franconian engineering arc with 20 years of Nuremberg-based experience, I am actively seeking re-immersion in the region's high-tech projects. I offer immediate availability and a workflow built around deterministic real-time execution—where prepared quantum states are cleanly injected into the circuit, and Ibex goats remain happily unentangled. My focus is on driving pro-technical economic expansion across the UK and EU, bridging regional innovation hubs with scalable, high-impact engineering interventions, as an engager and not to be confused by just a simple engineer. $\Psi \times$ State vector meets decision vector.

The Lineage: ... BCE → Psion → Symbian → Sendo → TI → ST-Ericsson → Infineon Comneon → Intel → TI → Dialog → TI → Infineon → Infineon → Renesas → TI (return arcs)

Key Fundamental Technical Areas

As an experienced engager, I offer broad expertise and practical intuition to every project.

- **Mixed-Signal & Power:** Expert in silicon bring-up, PDN, and high-speed mixed-signal validation. DC-DC converters, controllers, bi-directional designs, wearables/displays, PMICs, and BOOSTs. Extensive experience with post-silicon lab interventions, covering all measurements and test cases. Deep mixed-signal lab expertise at Infineon, TI, Dialog, and Renesas. Need I say more?
- **Physicochemical Systems:** Specialising in molecular spectroscopy and quantum-sensitive measurements, bridging the gap at the chemical–hardware interface. Possess extensive expertise in materials science characterisation, spanning the taxonomy of one-dimensional, two-dimensional, and three-dimensional materials and dielectrics. Particular emphasis is placed on understanding how crystal structure and dielectric properties influence system performance.
- **Electrochemical Energy & Materials:** Leveraging electrochemistry (EIS/DRT) for materials-aware battery system diagnostics and modelling. Battery Measurements, BMS and EIS/Diagnostics. Somewhat wetter fundamental chemistry and analytics.
- **Analytical Architecture:** Skilled in cross-domain system modelling and structured technical problem resolution.
- **Regulated Markets:** Extensive experience in MedTech and Scientific Research Equipment.

- **High-Precision Solid-State Sensing:** Expert in the characterisation and validation of solid-state optical front-ends, specifically the migration from legacy vacuum-based PMT glass to high-density Semiconductor SiPM and SPAD arrays; specializes in defining the deterministic performance limits of solid-state precision semiconductor sensing through rigorous measurement of dark-count rates, breakdown voltage stability, and photon detection efficiency (PDE).
- **Precision Optical Interfacing & Signal Conditioning:** Expert in architecting low-noise analogue front-ends (AFE) for discrete photon detection, specialising in high-fidelity signal conditioning from both legacy PMT glass and modern solid-state precision semiconductor sensing (SiPM/SPAD) arrays. Ensures deterministic event capture through meticulous management of impedance matching, high-speed transimpedance amplification (TIA), and thermal drift compensation in spectroscopic environments (NMR, IR, XRF, FIA, PAS, RAMAN, Microwave ESR/EPR). Possesses extensive capabilities in optical device characterisation, verification, and validation—skills gained from expertise in precision semiconductor and physical systems. Applies deep Chemistry or Physics and Electronics insights to these advanced systems.
- **RF Systems:** Broad experience in baseband and RF system environments, including 4G/5G, IoT, GNSS, LNAs, PLLs, VCOs, RF/TX chains, and I/Q demodulation. Developed and executed comprehensive systems-level test strategies, encompassing functional validation, interoperability testing, and performance benchmarking across diverse wireless standards and platforms. Skilled in designing automated test setups, troubleshooting complex integration issues, and ensuring rigorous compliance with industry specifications.
- **Cross-Domain Resolution:** High-level system modelling and structured problem resolution across disparate physical and digital domains, bridging the gap from ancient Greek foundations to modern tech execution.

Observing for the data-obsessed — capturing photons across the full electromagnetic spectrum, processing signals from DC and LF fundamentals through THz phenomena to scintillator-captured gamma pulses and beyond.

Shaping Electrochemical Energy-Storage Test Engineering through Fickian transport literacy, Goodenough-aligned intercalation insight, and Butler–Volmer kinetic mastery — delivering what BEREK promised: Batteries for a Longer Life! even though lithium begins degrading the moment formation ends.

Semiconductor production is inherently conservative: once a workflow is validated, it stays frozen. In FAB operations the principle is absolute — if it works, you leave it alone. Silicon delivery outranks every modernisation initiative, and this is why Perl, TCL, and legacy automation stacks remain in active use. They are not “old”; they are validated, deterministic components of a manufacturing chain where stability is the only metric that matters.

Seniority-as-a-Service

Strategic Technical Intervention | Seniority-as-a-Service I provide senior-level expertise to neutralise technical risk and accelerate hardware roadmaps without the onboarding lag of a permanent hire. With a 20-year track record across Tier-1 semiconductor companies, specialised MedTech, and UK Highways Infrastructure, I deliver deterministic solutions for mission-critical bandwidth bursts. Based in Central England, I am available for assignments across the EU, Greece, Ireland, and the DACH region. I can return to Germany after leaving in November 2025.

- **Immediate Impact:** Zero-ramp-up integration into complex BMS, Batteries/Electrochemistry, FPGA PDN, Semiconductor VVC/PVT workflows.
- **Risk Mitigation:** Expert-level structured problem resolution for high-stakes Mixed-Signal Electronics design Power Systems, Medical and Quantum Scientific Equipment.

- **Logistic Agility:** Dual UK/EU national; fully mobile across the Bavarian/Franconian arc and UK innovation hubs.
 - **Engagement:** Focused on delivering specific project outcomes via **Contract (hourly/day-rate)** or **Fixed-Term Contract (FTC)** specialist roles.
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Technical Credentials & Professional Development

- **MSc, Real-Time Systems** — embedded and real-time software/hardware systems
 - **BSc, Chemical Sciences** — electrochemistry, photochemistry, and materials science foundation
 - **HNC, Electrical & Electronics Engineering** — analogue, digital, and power electronics
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Specialist Short Courses & Summer Schools

- **RF Systems Summer School** — University of Surrey (RF design, propagation, MIMICs, Phase Noise, PLL, and systems-level engineering)
 - **Advanced Electrode Kinetics Summer School** — University of Southampton (electrochemical impedance, reaction kinetics, and electrode interface theory)
 - **NI TestStand, LabVIEW & PXI** — National Instruments certified training (automated test sequencing, data acquisition, and modular instrumentation)
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Software, Languages & Tools

- **Languages & Libraries:** Python/OOP (data analysis, test automation, instrumentation control) · MATLAB (signal processing, electrochemical modelling) · C# (hardware interface applications) · JavaScript (Obsidian dashboards) · Pandas & NumPy (numerical analysis, time-series) · Plotly (interactive measurement visualisation)
 - **Tools & Environments:** PyCharm · Visual Studio · PyTest · Jenkins · Docker · ARM Toolchains · NI TestStand · LabVIEW · NI PXI
 - **CAE & PCB Design:** Mentor Expedition · OrCAD · Altium Designer · PROTEL · SPICE · MATLAB Simulink
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Career Phase II — Contract Engagements: Semiconductor & Deep-Tech, NUE, MUC & DACH (2004–Present)

- **Contracting Partner:** Hays AG — primary vehicle for Tier-1 semiconductor and deep-tech engagements, selected for technical depth and relevance rather than simple availability.
- **Logicdev GmbH** (Remote / Graz) — Co-Founder & Chief Technical Adviser | 2020–Present. Unremunerated co-founder role running in parallel with contract engagements. Technical authority across battery and energy storage architecture, advanced BMS, AI/ML pipelines, and UAV hardware. Led Horizon EU bid development.
- **Texas Instruments (Freising & Aalborg)** — Multiple Contracts | 2004–2005, 2014–2020, 2023
 - *ISFET Characterisation Specialist (2023):* Built custom microfluidic MEMS wafer-prober stations for high-precision electrochemical sensing.
 - *Validation Expert (2018–2020):* Low-noise, dynamic validation for high-performance DC-DC stages; delivered automated characterisation workflows.

- *Validation Engineer (2014–2017)*: Precision analogue and mixed-signal validation; AFE and sensor signal-chain characterisation.
- *Baseband Engineer, Aalborg (2004–2005)*: Baseband hardware engineering for 2G/3G mobile platform development.
- **Infineon AG (Am Campeon)** — Multiple Contracts | 2020–2022
 - *Post-Silicon Verification Engineer (2021–2022)*: Automotive-grade SiC/IGBT gate-driver validation and FMEDA contributions for safety-critical systems.
 - *Validation Engineer for Radar Applications (2020–2021)*: 24GHz and 60GHz FMCW radar validation in Agile and Bare-Metal builds development environments.
- **Renesas Electronics / Dialog Semiconductor (Germering)** — Mixed-Signal Contracts | 2017–2018, 2022–2023
 - *Mixed-Signal Contractor, Renesas (2022–2023)*: Validation and triage for high-efficiency BOOST converters; improved silicon stability and reduced debug cycles.
 - *Evaluation & Validation Engineer, Dialog (2017–2018)*: Mixed-signal validation and silicon evaluation for power management ICs or Radar
- **Intel Mobile Communications / Comneon (Nürnberg)** — Baseband Contracts | 2010–2014
 - *Senior Baseband Digital Design Engineer, Intel (2011–2014)*: 2G/3G/4G platform verification and FAE support across global handset programmes.
 - *Senior Baseband Digital Design Engineer, Comneon (2010–2011)*: Baseband platform design and verification for 3G mobile communications.
- **Ericsson / ST-Ericsson (Nürnberg)** — Consultant Hardware Engineer | 2006–2009
 - Power systems on FPGA platforms; contributed to the first end-to-end LTE handheld call demonstrated at 3GSM Barcelona.

Career Phase I — UK Engineering Foundation | 1980s–2003

- **Overview**: Roles held across Oxford, Thames Valley, and Birmingham. Individual engagement dates available on request.
- **Highways Agency / SERCO (M25 MIDAS Subsystem)**: Acting Design Authority, Outstations. Infrastructure equipment approvals, site pre-commissioning, and technical support across PCO regions (London, Manchester, Leeds). Implemented 4 kV galvanically isolated RS485 links; designed equipment racks to Highways Agency functional requirements including TRG 1068 EMI and environmental performance standards.
- **Medical & Scientific Technology (Oxford Analytical Instruments, Oakfield, Resonance Instruments, Innotron Diagnostics)**: Developed safety-critical electronics for neonatal infusion pumps and gamma-camera organ-recovery post-op monitoring; designed wide-line TD NMR, 2D NMR benchtop, EDXRF, and FIA (Fluoroimmunoassay) instrumentation for industrial materials, foodstuffs, and clinical diagnostics analysis.
- **Traffic Systems (Golden River Traffic)**: Designed high-fidelity analogue front-ends for electrochemical sensors and vehicle-detection systems.
- **Mobile Platform Architecture (Psion / Sendo) — Design Lead**: Design Lead for ARM-based handhelds covering 2G/WLAN integration and mobile SAR compliance.
- **Project Chemist — Photoelectrochemistry (BEREC APG)**: Original research into photoelectrochemistry of binary and ternary chalcogenide semiconductor materials for photovoltaic screening — the foundation of the materials knowledge that runs through the rest of this CV.
- **Solar Systems Technologist (Venture Technology)**: Hardware development, characterisation, and installation for photovoltaic systems and power electronics — cell test, module assembly, battery chargers, and environmental radiation monitoring.